

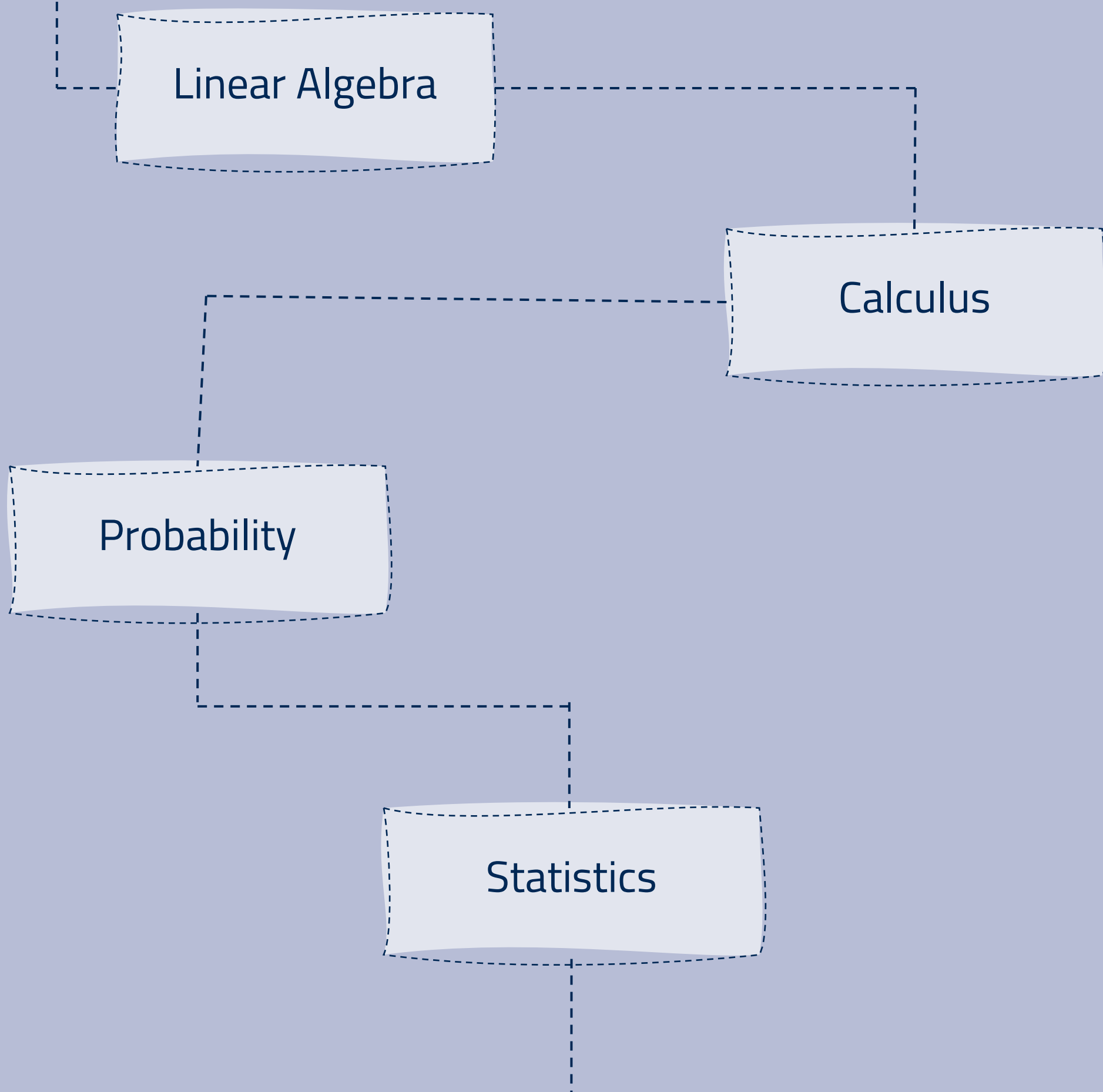
Data Science Track

Basic Math and Statistics for Data Science

Basic Math and Statistics for Data Science

Mathematics and statistics are at the core of data science, enabling us to analyze data, build models, and make informed decisions.

- This chapter covers essential mathematical concepts used in data science, including:



These topics form the foundation for machine learning, optimization, and predictive modeling.



Data scientist without
mathematics skills



Data scientist with
mathematics skills

Lets get started

Linear Algebra

• Vectors and Matrices

is fundamental in data science as it provides tools for:

- ✓ handling multi-dimensional data,
- ✓ transformations,
- ✓ and optimizations.

- Vector
A one-dimensional array (either a row or a column).
- Matrix
A two-dimensional array (rows $\times$ columns). Matrices store numerical data and are widely used in machine learning models.

Behind the science

The study of vectors and matrices traces back to ancient civilizations, but it was Muslim mathematicians who laid the groundwork for what we now call linear algebra. Muhammad ibn Mūsā al-Khwārizmī, a 9th-century Persian mathematician, made significant contributions to algebra (from the Arabic word "al-jabr"), which later influenced matrix theory. His works in solving linear equations set the foundation for modern matrix operations used today in science and engineering.

Al-Khwarizmi



Contributions to Humanity

Artificial Intelligence (Self-Driving Cars):

Matrices are integral in processing sensor data and making real-time decisions in autonomous vehicles.



PageRank



Did you know that

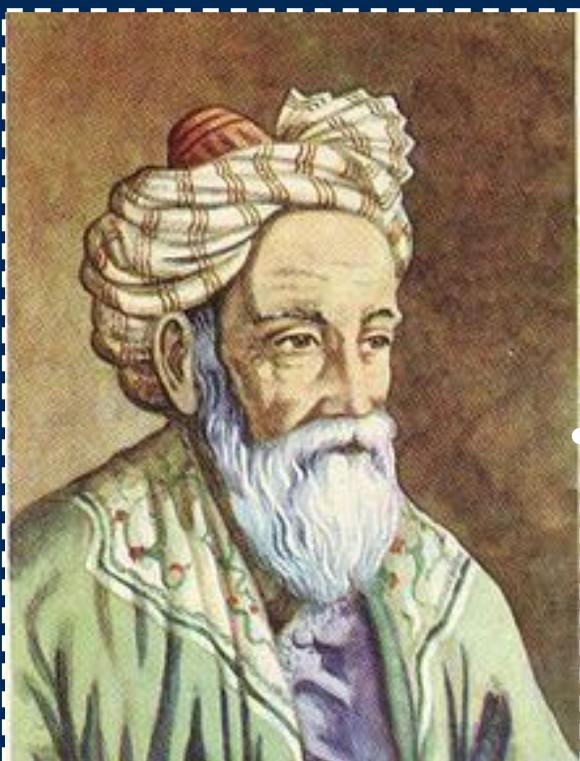
Google's search engine is powered by matrices? The PageRank algorithm, developed by Larry Page and Sergey Brin, uses matrix computations to rank web pages based on their importance. Without matrices, Google's search results wouldn't be as accurate or fast!

Applications in Data Science

Natural Language Processing (NLP): In NLP, matrices represent various data structures, including term-document matrices and word embeddings. Matrix factorization techniques are used in topic modeling and dimensionality reduction.



Basic Matrix Operations



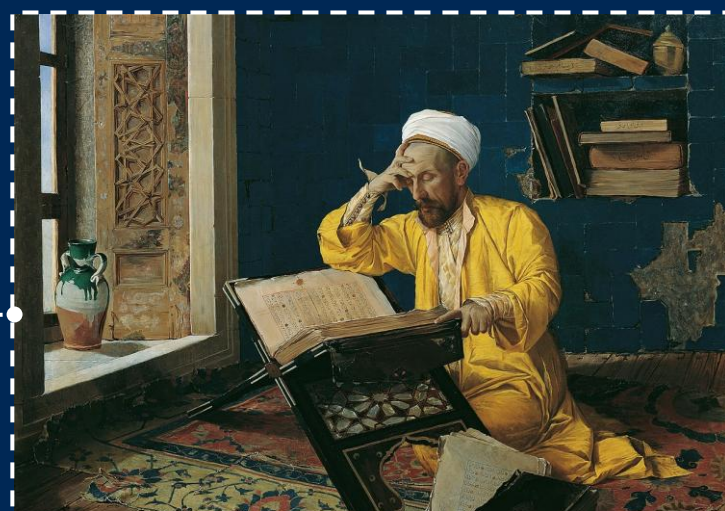
Omar Khayyam

Basic Matrix Operations

- Addition/Subtraction: Performed element-wise on matrices of the same size.
- Element-wise Multiplication: Each element of one matrix is multiplied by the corresponding element of another.
- Matrix Multiplication: Combines rows of one matrix with columns of another, used in transformations and linear regression.
- Inverse of a Matrix: If a matrix B is invertible, multiplying it with its inverse gives the identity matrix. Used in solving linear equations.

Behind the science

The systematic approach to matrix operations originated from the work of 16th-century mathematicians, but the concept of solving systems of equations with matrices has deeper roots in the Islamic Golden Age. Mathematicians like Al-Karaji and Omar Khayyam worked with equations that involved unknowns in tabular forms, an early representation of matrices.



Al-Karaji



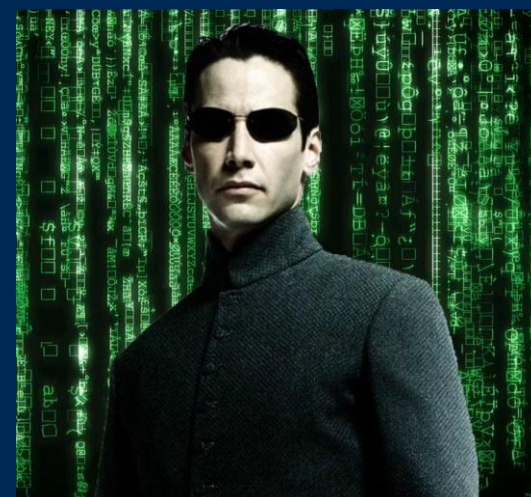
Contributions to Humanity

Cryptography:

Matrix operations, particularly matrix multiplication and inverses, are integral to encryption methods like the Hill cipher. This cipher uses a key matrix to encrypt messages, with its inverse used for decryption, highlighting the importance of matrix operations in secure communication.

Did you know that

In the movie "The Matrix", the falling green symbols are actually a mix of reversed characters from the Japanese katakana alphabet and mathematical symbols, hinting at the deep connection between matrices and computing.



Applications in Data Science

Matrix multiplication is the backbone of artificial neural networks, powering deep learning. From transforming data in recommendation systems (e.g., Netflix's movie suggestions) to performing singular value decomposition (SVD) in dimensionality reduction, matrix operations drive predictive analytics.



Dot Product vs. Cross Product



- Dot Product:
Computes a scalar value by summing the product of corresponding elements of two vectors.
- Cross Product:
Produces a new vector perpendicular to the two input vectors (only applicable in 3D).

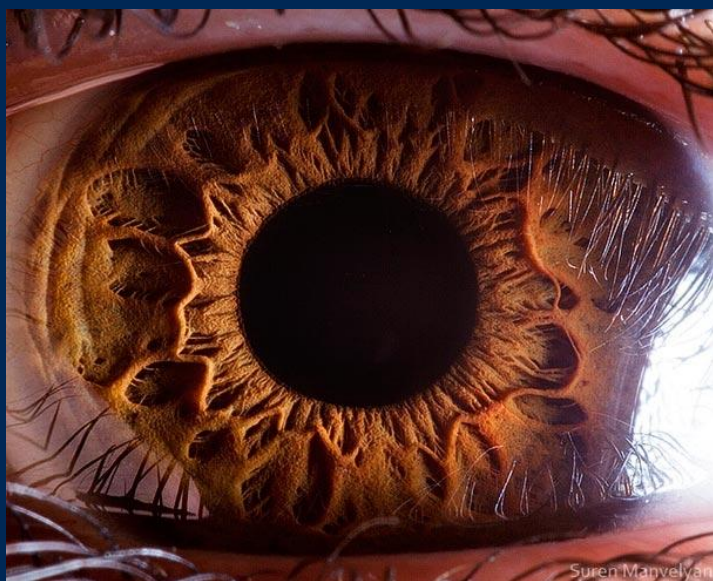
Behind the Science

The distinction between the dot product and cross product was formalized in the 19th century. However, Islamic mathematicians had already explored geometric and algebraic representations of vectors centuries earlier. Mathematicians like Abu'l-Wafa al-Buzjani studied geometric projections, a precursor to the dot product, and used these concepts in astronomy and engineering.

Contributions to Humanity

Space Missions:

Vector operations like the dot product and cross product are the unsung heroes behind space exploration! They calculate spacecraft trajectories, help with energy and work calculations, and ensure precise rotational control for missions. Without these, navigating the vastness of space would be impossible!



Fun Fact

Your eyes use the dot product! The way your brain perceives brightness from different angles is based on the dot product of light direction and the surface normal of an object. This explains why surfaces look brighter when light shines directly on them and darker when viewed at an angle.

Applications in Data Science

In 3D computer vision, the dot product calculates surface orientation for lighting and shading, while the cross product determines surface normals and rotational forces. Together, they enable realistic rendering and object recognition in AR, VR, and robotics. These concepts are crucial for analyzing and simulating 3D environments.



Calculus

Functions, Derivatives, and Integration.

Calculus is used in data science to optimize models, measure change, and compute gradients for learning algorithms.

Definition:
Measures the rate of change of a function at a given point.

Differentiation

Gradient (Slope):
Helps in gradient descent, a key optimization method in machine learning.

For example, if
 $f(x) = x^2 + 3x + 2$
its derivative is:
 $f'(x) = 2x + 3$

This represents the slope of the function at any given x .

Integration

Calculates the area under a curve.

Indefinite Integra

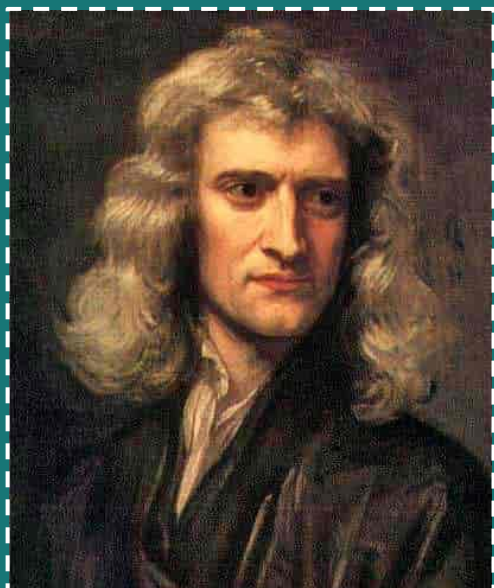
Represents the general form of accumulated values.

Definite Integral

Measures the total accumulation between two points.

$$\int (x^2 + 3x + 2) dx = \frac{x^3}{3} + \frac{3x^2}{2} + 2x + c$$

Integration is used in probability distributions and deep learning optimizations.



Isaac Newton

Isaac Newton was born in 1642, the year Galileo passed away. When he joined Cambridge University in 1661, he had little mathematical knowledge but quickly advanced by studying Euclid and Descartes and attending Isaac Barrow's lectures. In 1665 and 1666, Cambridge closed due to the plague, and Newton returned home. During this period, he made four groundbreaking discoveries: (1) representing functions as infinite series, including the binomial theorem, (2) developing differential and integral calculus, (3) formulating the laws of motion and universal gravitation, and (4) conducting prism experiments that revealed the nature of light and color.

Contributions to Humanity



Theme Park Line Optimization (Disney & Universal Studios)

- Ever noticed how Disney's FastPass or Universal's Virtual Queue somehow reduces wait times while keeping the park from feeling overcrowded?
- Calculus-based queue optimization models analyze visitor flow, predicting which rides will have the longest lines at different times of the day.
- By adjusting reservation slots dynamically, theme parks balance the guest experience while maximizing ride capacity.

Funny Fact

The Ice Cream Cone Paradox

Here's something funny and real: In the world of calculus, you could theoretically calculate how an ice cream cone melts! If you apply heat diffusion equations (thank you, Fourier!), you can figure out how fast your ice cream would melt and where it drips—before it even starts!

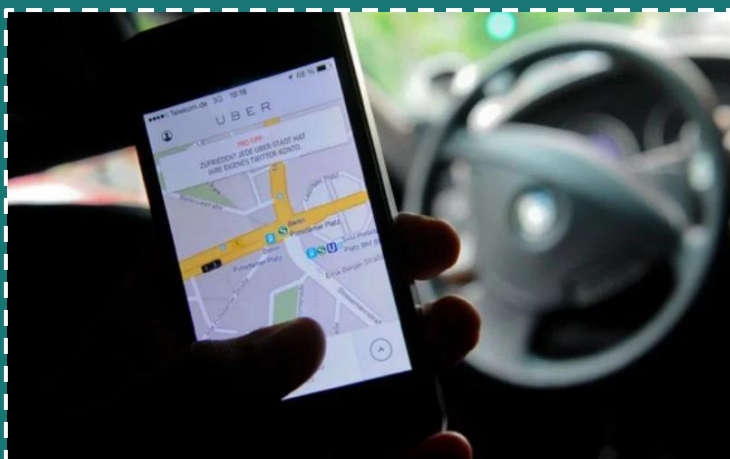
So, next time your cone is dripping, just remember: math might be the real reason your ice cream's running away from you!



Applications in Data Science

Ride-Sharing Apps (Uber, Lyft, Careem)

- Ever booked an Uber and seen how the ETA updates constantly? That's calculus in action!
- Uber predicts travel time using differentiation (to calculate rate of speed changes) and integration (to estimate total time over a route).
- It even adjusts prices dynamically based on supply-demand curves—a direct application of calculus-based optimization.



Probability

Probability is the study of uncertainty and likelihood, playing a crucial role in predictive modeling and statistical inference.

Basic Probability

The probability of an event E occurring is:
 $P(E) = \text{Number of favorable outcomes} / \text{Total number of outcomes}$

For example, rolling a six-sided die, the probability of getting a 5 is: $P(5) = \frac{1}{6}$

Behind the Science

The Origins of Probability Theory

The formal development of probability theory began in the 17th century with the correspondence between French mathematicians Blaise Pascal and Pierre de Fermat. In 1654, they discussed problems related to games of chance, leading to foundational principles of probability. Their work was further advanced by Christiaan Huygens, who published "De Ratiociniis in Ludo Aleae" in 1657, the first book on mathematical probability.



Pascal

Fermat

Compound Events



Independent Events

The occurrence of one event does not affect the other.

Example: Rolling a die and flipping a coin.

Dependent Events

The outcome of one event influences the other.

Example: Drawing cards without replacement.



Mutually Exclusive Events

Events that cannot happen together

Example: rolling a 3 and a 5 at the same time

Fun Fact

A Curious Historical Anecdote

In the 18th century, the term "chance" was used in the mathematical sense of "probability," and probability theory was referred to as the "Doctrine of Chances." This terminology reflects the evolving understanding of randomness and uncertainty during that period.



Complementary Events

The probability of an event not occurring is:
 $P(E') = 1 - P(E)$

If the probability of rolling a 4 is $\frac{1}{6}$, then the probability of not rolling a 4 is:

$$P(\text{not } 4) = 1 - \frac{1}{6} = \frac{5}{6}$$

Contributions to Humanity

Weather Forecasting

Meteorologists use probability models to predict weather conditions, such as the likelihood of precipitation, aiding individuals and organizations in planning activities.



Applications in Data Science

Risk Assessment

Data scientists employ probabilistic models to evaluate uncertainties and potential risks in projects, enabling informed decision-making.

Statistics

Statistics is divided into descriptive and inferential statistics, both essential for analyzing data.

Descriptive Statistics

Descriptive statistics summarize and describe data using measures of central tendency and variability.

Measures of Central Tendency

- Mean (μ)
The average of all values.

$$\mu = \frac{\sum x_i}{n}$$

- Mode
The most frequently occurring value.
- Median
The middle value when sorted.

Behind the Science

In the early 20th century, Sir Ronald A. Fisher revolutionized the field by developing the analysis of variance (ANOVA) and experimental design, establishing statistics as a rigorous scientific discipline.



Measures of Variability

- **Range**
Difference between the maximum and minimum values.

- **Standard Deviation (σ)**
The square root of variance, indicating data dispersion.

- **Variance (σ^2)**
Measures the spread of data around the mean.

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{n}$$

Fun Fact

A Curious Historical Anecdote

In the early 20th century, statistician Ladislaus Bortkiewicz analyzed data from the Prussian army regarding deaths caused by horse kicks. He found that these rare events followed a Poisson distribution.



Measures of Spread

- **Five-Number Summary**
Consists of Minimum, Q_1 (25th percentile), Median (Q_2), Q_3 (75th percentile), and Maximum.

- **Interquartile Range (IQR)**

$$IQR = Q_3 - Q_1$$

Inferential Statistics

Inferential statistics help us make predictions about a population based on a sample.

Confidence Intervals

Provide a range in which the true population parameter is expected to fall.

Hypothesis Testing

Determines whether an observed effect is statistically significant.

Regression Analysis

Models relationships between variables.

Contributions to Humanity

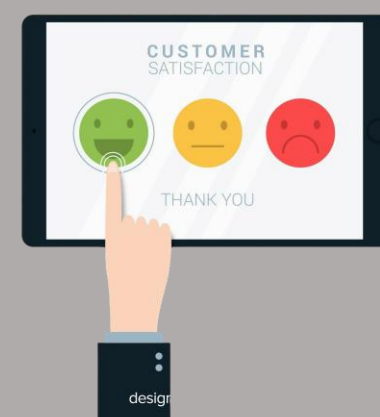
Policy Making

Governments rely on statistical data to inform policies on education, healthcare, and economic planning, ensuring decisions are based on empirical evidence.



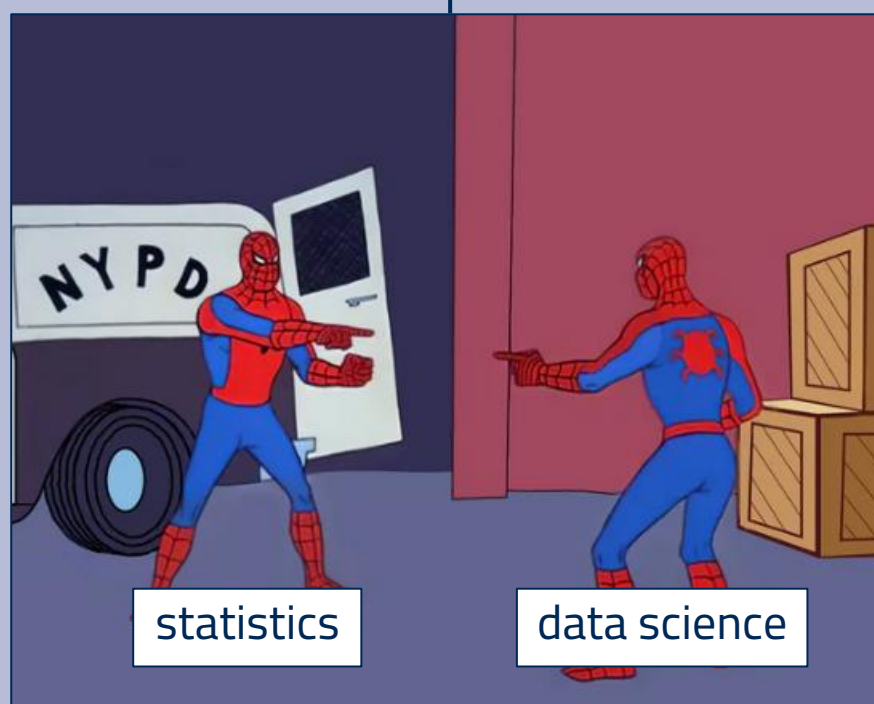
Applications in Data Science

Customer Churn Prediction in Telecom Industry
predictive modeling is extensively applied in industries like telecom to predict customer churn. Statistical methods, such as regression analysis, enable companies to identify customers who are likely to leave, allowing them to take proactive steps to retain them.



Conclusion

Mathematics and statistics are fundamental to data science, forming the basis for data analysis, machine learning, and artificial intelligence.



Mastering these concepts enables data scientists to build accurate models, make informed decisions, and optimize algorithms efficiently.

